JMB Lighting — User Manual

JMB Zone / 4our / 8ight · Hardware 3.0.0 · 2026-10-31 · 4-port shown (1/8-port variants identical unless noted)

1. Front panel

One rotary encoder: **turn** = scroll / adjust, **click** = select, **hold (1.2 s)** = back one level, **keep holding (3 s)** = straight home + lock from any depth. The home screen shows the JMB mark; hold to unlock into the menu. The panel relocks after inactivity (running tests and effects keep going). When no source is decoding, the lock screen reads **No Data** — whatever the CRMX link state — so a linked-but-silent radio can never masquerade as a live source.

Main menu

Item	What it does
Test Mode	Step / Fade output tests (click cycles, hold exits)
Ports	per-port or All Ports setup, incl. **Presets** — see below
Sources	**the input hub** — wired DMX and CRMX each with live status (the active source + Hz shows inline and in the header); Universe Listen arms here
Health	worst fault inline; Clear Faults + the live universe viewer
Setup	system only: ID, Bluetooth + PIN, DMX hold, curve (all ports), encoder, language, protection, cal, reset

Port Settings (per port, or All Ports)

The rows use the app's words, so one word means one thing on both: **Address · Head · Mode · Manual · Effect · Speed · Curve · PWM · WB · Limit · Trip · Clear Fault · Presets · Exit.**

Address and Head open the digit editor (turn = digit, click = next, SAVE commits; hold cancels; the footprint hint shows the span). Mode cycles the list in the DMX chart's order — Dim 1ch / Dim 2ch / CW/WW 8 / CW/WW 16 / RGB 8 / RGB 16 / **RGB FX** / RGBW 8 / RGBW 16 / **RGBW FX** / RGBWA 8 / RGBWA 16 / **RGBWA FX**; on All Ports every port takes the same mode. **Manual** On/Off: Off hands the port back to DMX and stops any effect; the colours are kept for the next On (the same thing as Presets → DMX In). Effect opens the effect picker; Speed is the 0–100 % scrub editor. **Curve** is this port's dimmer curve (Lin / Soft 1.6 / Thtr 2.2). PWM cycles the PWM library: **Smooth 8 kHz / Silent 20 kHz / Camera 15.6 kHz**, then any custom entry that has been set in the app (shown as **C1 12.3k**; see §8). **WB** opens the white-balance editor: one row per output the mode drives (R G B, plus W on RGBW and W A on RGBWA; Cool / Warm on CW/WW), turn = 0–255 trim, click = next row, then **Reset** (no trim anywhere) and **Save**; the port lights at the preview level while the editor is open and drops back on Save or a hold (cancel). **Limit** steps the current trip from 0.25 A in 0.25 A steps to 2 A and 0.5 A steps to the cap; **Trip** steps the temperature trip 40 → 110 °C in 5 °C steps; both reach the port hardware immediately. Clear Fault re-arms a latched port. **Presets** opens the presets sub-menu (below).

All Ports adds five rows: - **Universe: Local / Send** — Local edits this unit. **Send** also passes a baton down the DMX line to every unit armed to Listen after you commit (see §4) — addresses, the dimmer master, a look, or the white balance (**SEND WB?**). "Universe" is the app's word for those units. - **Apply: Same / Seq / Offset** — how a base address fans out: everyone the same; follow-on by each port's real footprint; or a fixed 14-channel spacing, the widest mode (RGBWA FX), so any port can change mode later without re-patching. The app, the panel, a desk's RDM base address and the chain baton all use the same rule. - **Tempo** (60–240 bpm) and **Spread** (0–100 %) — the effects clock and the port-to-port offset, live, the same as the app's Effects card. - **Clear All** — every port to dimmer 0, manual on, effect off, colour 100/100/100, speed 128 (the app's Clear all).

Presets (Port Settings → Presets)

Three rows: - **Dimmer** — a live master brightness. On a single port it opens straight on that port; on **All Ports** it walks each port in turn (turn = level, click = next port). The output follows the encoder **live** — no need to confirm to see it. 1 % steps on a slow turn, coarser when you spin. Hold = cancel and restore the levels you started with. - **Colours** — the factory colour / CCT looks. - **Scenes** — the controller's stored scenes, the same list the app's Scenes tab shows (up to 100). **Follow On/Off** steps through them (each holds for its fade + wait, round and round); click a scene to recall it with its fade; **Save new scene** captures the rig's look as "Scene k" (rename, retime, reorder and delete in the app).

When Port Settings is on **All Ports** *and* **Universe: Send**, the Dimmer row becomes a **Universe Master**: you set one level and, after a **SEND DIMMER?** confirm, it sets every port on this unit *and* every **armed** unit down the DMX line (see §4). Send from the **first** controller in the line to reach the whole rig.

Sources (the input hub)

One screen answers "what's driving the rig": the header names the **active source and its refresh rate**, and each row carries live status — **DMX In** (Hz or *silent*; click = the line analyser), **CRMX** (*no link / NO DATA /* Hz — data-driven, so "linked but nothing arriving" is visible at a glance; click = link/unlink; the external antenna is always on) and **Universe Listen** (arms this unit to accept ONE baton from up the line — always Off at power-up, disarms itself after use; the app's Setup → Universe → Listen is the same switch).

Setup

Controller ID (shows on screens, in the Bluetooth name after reboot, and in the RDM label) · Bluetooth on/off + PIN · **DMX Hold** (how many seconds the last look is kept after the signal is lost, 0–60; 0 = forever) · Curve (all) (sets every port's curve at once; the per-port row is Ports → Curve) · Reverse Encoder · Language · Current Limit + Limit Lock · Current Fold · Thermal Fold · Calibrate temp · Factory Reset (confirmed; it unlinks the CRMX receiver first, then erases and restarts). Bluetooth can be switched off from here on any unit (it ships on); switch it back on the same way.

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2. Output priority

Test mode → Identify (RDM/app) → live DMX → Effect → Manual → hold-last. Live DMX always outranks manual and effects; when the desk vanishes, the last DMX look holds until you set something

else. At power-up the outputs stay dark for just over a second, so the supply's inrush bypass has closed before the strips draw current, and then fade up over half a second. All sources pass the dimmer curve.

"Live DMX" is the wired input *or* the CRMX radio — they merge into one universe, so everything downstream (patch, FX personality, the diagnostics viewer, RDM) treats the radio exactly like a cable. The header on every screen names the winner: **DMX** or **CRMX**.

DMX thru is active, not a loop-through. The OUT socket is re-driven by its own line transceiver, so the signal leaving the unit is a fresh copy of what arrived (bit-for-bit, no software in the path) and a long daisy-chain stays clean however many units are in it. Downstream fixtures' RDM answers are relayed back up to the desk through the same path.

2a. Firmware update over Bluetooth (BLE OTA)

The controller updates its own firmware straight over Bluetooth from the app — no USB, no cable.

- **One button: Update Firmware** (Setup → *Firmware update*). It is only offered when there is actually something to install — otherwise it reads *Firmware up to date*. When an update is waiting, the unlock screen says so as you connect, so you find out before going looking. It updates everything the unit runs, in one go. Outputs go dark for a few seconds part way through, and the controller restarts into its new firmware at the end. Nothing needs choosing and nothing needs finding — the firmware is published alongside the app. The firmware is published alongside the app, so there is no file to find and no way to send a build meant for a different product. The card names the version and build about to be installed, and the build the controller is running, so you can see whether an update would change anything.
- **Only what differs is installed.** The controller and its port module each have their own image; the run installs the one(s) that have changed and leaves a port module already on the published image alone, so a controller-only build is a controller-only update. If the button reads *Update port module*, that is all it will touch.
- **Checked twice before it runs.** The app verifies the download's size and checksum *before* opening a transfer, and the controller re-checks the checksum across the Bluetooth link before the image is made bootable. A bad download fails in a second rather than after a long upload.
- **Automatic rollback.** The new image boots on probation: if it doesn't come up cleanly and confirm itself, the controller returns to the previous working firmware on its own. The app shows *confirming…* during probation and *firmware confirmed* once the build has proven itself.
- **Manual rollback.** *Roll back* returns everything to the previous firmware deliberately — for a build that runs perfectly well but behaves wrongly, which the automatic check cannot detect. The button stays greyed until there is a previous version stored, which happens after your first update.
- **Do not power the unit off during an update.** If one fails part way, the unit keeps the firmware it already had — nothing is committed until it has been checked — and it can simply be retried over Bluetooth. If the app reports *a stale copy is still being served*, the download site has not yet caught up with a very recent publish: wait a few minutes and try again.
- **Controller only** (Developer mode, §6 Setup) skips a pending port-module step. It exists for one case — a port step that drops the link every time — and is otherwise unnecessary.

> Keep the app open and the phone unlocked until an update finishes. Outputs hold their last look during a controller update; they go dark during a port module update. The app's other controls wait until the update has finished.

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3. Effects

Twenty-one built-in dynamic looks per port (Candle, Fire, Fireworks, Water, Siren, Laser, Club, Lightning, Rainbow, Aurora, Sunset, TV Sim, Strobe, Police, Pulse, Party, Traffic, **Fluoro, Paparazzi, Explosion, Clouds**). Set from Port Settings → Effect, the app, or DMX channel 8 of the FX personality. Speed 0–255 (128 = designed tempo) from the app or DMX channel 9. While an effect runs, the app's **RGB sliders are greyed out** — the effect owns the colour; clear the effect to get them back (the dimmer and speed wheels stay live). **Speed 0 stops an effect**: a continuous look freezes on its current frame, Strobe and Police go dark. **Strobe and Police are clocked on the port module itself** — the controller sends the flash pattern once and the port keeps time on its own clock, so the rate is exact at every speed and identical across every unit running it. The top of the Strobe speed wheel is compressed: above the mid-point the rate rises more gently, and 255 is about 30 flashes a second, the fastest the eye still reads as a strobe. **Effects never start on their own.** When DMX is absent, an effect runs only from the moment you deliberately pick it (panel or app) — a saved effect does not auto-run at power-up, and a desk dropout (cable or CRMX) always **holds the last DMX look** rather than starting one. Pick the effect again after a loss to run it. Siren/Police alternate across ports like a lightbar; Party pulses fully-saturated colour on a 128 BPM bar clock; Fireworks bursts out of true darkness; **Sunset is a one-shot arc** — from golden hour it eases through ember red down to black over ~2 minutes and stays there; re-select it (or dip the port dimmer to 0 and back up) to run another sunset. Traffic plays passing-headlight sweeps with independent timing per port. **The film gags:** Fluoro is a failing fluorescent tube — rapid failed strikes, then it catches and holds with a faint mains buzz; Paparazzi fires irregular hard white pops, independent per port, like a press pit; Explosion is a white-hot flash blooming to an orange fireball with an ember-flicker decay; Clouds drifts daylight slowly dimmer and cooler as cloud crosses the sun, then back.

Rig-wide effect controls (the app's Tempo pop-up, from the Tempo button at the top of the Control tab): **Tap tempo** sets the whole rig's beat-effect speed to the music — tap the app's Tap button in time (or type a BPM, 60–240; 128 is the designed tempo). **Spread** offsets each port in time (0–100 %) so effects sweep across the rig as a wave instead of every port in lockstep. Both persist across reboots.

Capture to scene: the app's Scenes tab saves the current look — **Save scene…** (named) or **Capture live** (one tap, auto-named) — and the panel's Presets → Scenes → Save new scene does the same. Both keep the same thing, **in the controller**: each port's colour (white and amber too), level, effect and effect rate. An effect is kept as a setting, so it runs again from the start on recall; tempo and spread are rig-wide and not part of a scene. A scene recalls with a crossfade from either surface and rides in the show file. **Up to 100 scenes per phone** (they live in the phone's browser storage, and in any show file you export).

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4. The Universe (one screen addresses every unit down the line)

"Universe" is the set of units armed to **Listen** down the DMX line — the app's word, used on the panel too. To address them all from one screen:

1. On **every unit that should take part**: Sources → **Universe Listen** → **ARMED** (or the app's Setup → Universe → Listen). Units left Off are untouched — preset rigs are safe.
2. On unit 1: Port Settings → All Ports → Universe: **Send**, choose Apply mode, open Address, enter the base, SAVE.
3. Confirm the **SEND PATCH?** warning. Unit 1 addresses itself, then each armed unit down the line takes the follow-on addresses **and the next controller ID**, shows a receipt, disarms, and relays onward.

Head numbers travel the same way. A unit accepts one baton per arming. The app's Patch sheet does the same with its **Universe** button, and can send a **Mode** with the addresses.

The dimmer travels too. With the same arming in place, Port Settings → All Ports → Universe: Send → **Presets** → **Dimmer** gives a single **Universe Master** level. Set it, confirm **SEND DIMMER?**, and every armed unit down the line takes the same brightness. Only a downstream unit hears the baton, so send from the first controller in the line. (Addresses are untouched — this only moves the dimmer.)

Looks, blackout and full too. All Ports → Universe: Send → **Presets** → **Colours** → pick a colour/CCT and confirm **SEND LOOK?** to set that look on every armed unit. Choosing **Off** blacks out the whole universe; **White** takes it to full. The app's **Universe** section (Setup, under CRMX) has one-tap **Blackout / Full / DMX In** buttons for the same thing, and **Send settings** sends this unit's settings to every armed unit: curve, PWM, white balance, limit and trip per port, the controller's current limit and lock, folds, DMX hold, Apply mode, tempo and spread, and the custom PWM entries; never addresses, modes, heads, numbers, names, labels, the PIN or the custom presets. A unit with more ports than the sender gives its extra ports the sender's last port's settings, and each receiving unit shows **Universe settings set**. A port's white balance sheet has **Universe** to send its trims down the line the same way, and the panel's WB editor on All Ports with Universe: Send does the same (**SEND WB?**). (Dimmer, looks and white balance never re-address or renumber — they only change the output.)

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5. RDM

The controller is a full E1.20 responder: discovery, identify (flashes the ports and strobes the main screen **HERE**), device info, start address, personality — the root device carries the rack view (a SET fans out per the Apply mode) and each port is addressable individually as a sub-device. E1.37-1 dimmer curve and PWM frequency are settable from the desk, on the root for the whole rack and on each port. **White balance** (one parameter per output, raw 0–255: red, green, blue, white and amber), its **preview**, and the PWM library's **custom entries** are manufacturer parameters that a console can list and edit; the table is in the DMX chart. The controller also publishes **per-port temperature and current as E1.20 sensors** on the root device ("Port 1 Temp", "Port 1 Current", …), so any RDM console reads fixture health live. The main screen also flashes the moment discovery finds the unit. The powered DMX thru forwards everything downstream; downstream fixtures' RDM responses are relayed back to the desk automatically.

RDM controller (the app's Diagnostics tab). The controller is also an RDM *controller* for everything plugged into its thru line — a full RDM tester built into the rack. **Scan** discovers every downstream RDM fixture and lists it as a card: model, label, UID, start address, footprint, personality and sensor count. Tap a

card to edit it in place — **identify** (flash the fixture to find it), **start address**, **personality**, **label**, and live **sensor reads**. Scanning takes over the output line while it runs, so with an upstream desk live the app asks before taking over; the session releases the thru automatically when you stop, when you leave the Diagnostics tab or put the app away (any fixture still identifying is switched off on the way out), or **3 minutes after your last command** — the app then reads *stopped — idle for 3 minutes, thru released*, so a scan left open at a gig never holds the rig.

> **Discovery flashes fixtures.** When a desk (or this app) runs RDM discovery, every fixture it finds — this controller's ports included — flashes its output at full for a moment, and E1.20 *Identify* flashes it until switched off. That is the standard behaviour every RDM device shares; it is not the same as the app's *Flash to find*, which never touches an output (§6 Setup).

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6. The app

https://app.jmblighting.co.uk/ — runs in **Chrome or Edge** on Android or a desktop (on **iPhone**, open it in the **Bluefy** browser, which supplies the Bluetooth that iOS Safari lacks). On Android it is also on **Google Play** as *JMB Lighting*; otherwise choose **Add to Home screen** for a full-screen, offline-capable app. Tap **Connect** and pick your controller from the list. The factory PIN (**0123**) is tried for you, so a new controller opens straight away. A controller with its own PIN (set in Setup → Change PIN) asks for it once per phone, then remembers it for that controller only — a PIN typed for one unit is never offered to another. After five wrong PINs the controller ignores every PIN, the right one too, for 30 seconds, doubling each time up to 5 minutes; from the second miss the app says so, so wait before trying again. **Disconnect** under the PIN field backs out to the connect screen. If a controller has just been reflashed, toggle the phone's Bluetooth off and on once first. The pairing list shows each controller by its **name and ID** (Setup → *Controller and Bluetooth name* / ID; the Bluetooth name changes at the next restart), so give every unit its own before a multi-unit job; two Zones with the factory name look identical in that list.

Once connected, **tap the Bluetooth mark in the header** for the connection sheet: which controller this is, its Bluetooth name and port count, the link quality and firmware, and the ways out — **Flash to find**, **Connect to another controller** (drops this link and reopens the chooser) and **Disconnect**.

Everything the app sends goes **straight to the controller over Bluetooth** — no account, no cloud, no internet needed once it has loaded, and nothing about your rig leaves the phone. The app also **updates itself**: opened with any connection it loads the newest version, then keeps working from its cached copy at a venue. The Google Play app is the same app and updates the same way, so there are no updates to chase. **Open it once with a connection after installing it**; from then on it starts with no signal at all, aeroplane mode included, and on a venue Wi-Fi with no internet behind it, it starts from its saved copy within a few seconds.

Control — the working screen

One card per output. Each card holds the port's **DMX address** and **head number**, its **personality** (RGB, RGBW or RGBWA, each 8-bit, 16-bit or with the effect channels; tuneable white 8- or 16-bit; or a **single-channel dimmer** — one DMX level over the port's set colour, ideal for single-colour and IR strips), an

effect picker with a **speed** control, and the level controls: one slider per output the personality uses — **R / G / B**, plus **White** and **Amber** on the five-channel personalities, or **Cool / Warm** on a tuneable-white port (ice-blue and amber tracks) — plus two **encoders** for **Dim** and **FX speed** that are easy to ride by feel during a take. Under the encoders every port prints its settings in three columns, each headed by a button that opens the sheet to change them: **Patch** (Mode, Address, Head — the sheet also holds the **Label**, up to 16 characters, which shows in the card header and on that port's own screen when the panel is locked. Port 1's sheet also has the **Apply** row: pick Same / Seq / Offset, then **Ports 2–4** (2–8 on an 8ight) patches the other ports from port 1's address, or **Universe** does the same and sends the patch on down the DMX line to every unit armed to Listen, each following on from the last. A pop-up asks **DMX only** or **DMX + Mode** first, then the sheet closes and a message confirms what was set. Seq follows on by each port's footprint, Offset by a fixed 14-channel stride (the widest mode, RGBWA FX), heads follow +1; the panel's All Ports edit and a desk's RDM base address fan out by the same mode), **Output** (Curve, PWM, white balance) and **Protection** (Limit, Trip, faults). **Clear all** returns every port to a clean base — full white, effects off, dimmer down — after a confirmation. **Undo** puts the looks back — each port's colour, level and effect — on the controller they were cleared from; neither touches the patch.

A port whose readings stop arriving shows **no data** in red on its badge instead of passing old readings off as live (Telemetry also greys its figures); it clears as soon as readings return.

Under the header sits a row of **pills**: **DMX**, **CRMX**, **Test**, **ID**, **Cfold**, **Tfold** and **Fault**. They are always there; a pill is dark until its state is active and lights when it is. The first three say what is driving the outputs: DMX green while a signal drives the rig, CRMX purple while the radio carries it, Test amber while a test sequence overrides everything or the DMX transmitter (Diagnostics) is sending. ID is the controller's number. The last three say what is holding the outputs back or wrong: Cfold red while the current fold is scaling them and Tfold orange while the thermal fold is, each with its scale, and Fault red while any port reports a fault. Tap any pill, lit or dark, and a box opens under the row in that pill's own colour, saying what it means and where things stand; tap the pill again or the box to close it. Where there is something to do the box carries the button: **Stop test** (or **Stop transmitter**) on Test, **Go to port** on Fault. The frame rate is in Diagnostics. The pills stay pinned under the header on every tab, so they never move as a page scrolls. On the Control tab the buttons beneath them (**Clear all**, **Undo**, **Tempo**, **Reset WB**) stay pinned as well, so they are always in reach while the port cards scroll.

An **orange outline on a DMX address box** means that port's channels overlap another port's: its start address plus footprint runs into a different start address on the same controller. It is a nudge, not a block — overlapping can be deliberate — and both sides of the overlap are marked. Two ports set to the *same* start address are treated as mirroring and are not marked.

Touching any colour or level control **takes manual control automatically** and stops a running effect on that port, so a fader never feels dead. Pull every colour to zero and the port goes dark: the app takes its dimmer to 0 and keeps your level on the Dim encoder, and it comes back when you raise a colour. When a live **DMX or CRMX** signal is driving the rig, a **"Live DMX in control"** banner appears — your manual moves here are overridden until that signal stops, which is *why* the faders would otherwise seem to do nothing.

Output holds the **PWM mode** (the dimming carrier frequency — see §8), the **dimmer curve** (**Linear / Soft / Theatre** — how fader travel maps to brightness; a custom PWM entry brings its own and locks it) and **white balance** (below). **Protection** holds **Limit**, the current the output may draw (0.25 A up to **6.75 A**),

Trip, the temperature it shuts down at, and a **Clear fault & re-enable** button that appears only while the port is latched. A latched port shows its fault in red on the card and on the header badge; tapping the badge opens the sheet.

White balance. In the Output sheet, under the curve and PWM, is a master for each output the port's mode drives: Red, Green and Blue, plus White on RGBW and White and Amber on RGBWA, Cool and Warm on a tuneable-white port, one Level on a dimmer port. Drag the slider or type the value; 255 is full. The trim is applied after the dimmer curve, so the balance holds from full to the bottom of a fade and does not move if you change the curve. The **Preview** slider sets how bright the port shows while you adjust: the preview starts the moment you touch a trim, never over live DMX (with a desk connected, bring the channels up at the desk), and the port returns to its look when you close the sheet, leave the app or lose the link. A port with a trim in force shows an orange **WB** badge. **All ports** copies this port's trims to every port on the controller; **Universe** sends them to this unit and every armed unit down the DMX line, arming each unit first as in §4; **Reset** clears this port, and **Reset WB**, at the end of the row of buttons at the top of the Control tab, clears every port. The trims live in the controller and apply to every source.

Colour — dial a look, don't push channels

A console-style picker for the selected ports (choose **All** or one port at the top). Its five columns — **Dim, Hue, Sat, Kelvin** and **ΔUV** — are all views of one colour: nudge Hue/Sat and Kelvin follows; nudge Kelvin and Hue/Sat follows; the last column you touch wins. **ΔUV** is an independent **green/magenta tint** trim for matching a practical or pleasing a camera. On a tuneable-white port the picker becomes **Kelvin-only** with a live warm/cool mix readout. Reach for this when you want to *choose* a colour rather than balance raw R/G/B by hand.

Scenes — save and recall looks

Store the whole rig's current look as a **scene** and bring it back later, each with its own **crossfade time** (0 = snap). **Capture live** snapshots whatever the controller is outputting right now, auto-named by the time. **Scenes live in the controller** — the app is its remote — so the same list is on the unit's own panel (Ports → Presets → Scenes), plays back from either, and is still there when a different phone connects; the exported show file carries them too. Tap a scene's name to rename it; hold the name to lift the scene and drag it up or down into a new order. Tick the box at the left of a scene to pick it: **Delete** at the top removes every ticked scene (after a confirm), and dragging a ticked scene moves all the ticked ones together as a block. **The limit is 100 scenes per controller**; the app says so when it is reached, so delete a few before saving more.

Each scene has two times, in seconds, under the **Wait** and **Fade** headings: Fade is the crossfade into the scene, Wait is how long it holds after that fade before the list moves on. The buttons along the foot of the tab run the list, and the controller does the running: **Go** recalls the next scene (the first when none is lit, the first again after the last). **Cue** steps one scene per Go press; **Follow** carries on by itself, each scene holding for its Wait, round and round, until **Stop** — from the app or the panel's Follow row — and it keeps going if the phone drops away. Recall on a row jumps to that scene, and Go continues from there. Follow is always off after a power-up.

Telemetry — the health glance

A read-only status page — the same telemetry the port screens show, gathered in one place: each port's **patch**, **output current** (calibrated to within about 1 % of a bench meter), **temperature**, any **fault**, and its active **source**. Your quick "is everything healthy and drawing what I expect?" check. A port whose readings have stopped arriving reads **no data**, its figures greyed.

Diagnostics — the built-in DMX analyser

This is where the controller doubles as a **test instrument** for the signal on the wire. Each tool answers a specific question:

- **Signal card** — *Is a signal arriving, and is it healthy?* Shows whether a signal is present and from where (**DMX / CRMX**). The controller listens to both: the cable wins whenever it carries DMX, and the radio takes over when the cable goes quiet. It also shows the signal's **frame rate** with a rolling 90-second graph, **slot count**, **start code**, **time since the last packet**, alternate-start-code packet count, and **error counts**. Your first check that the desk is actually reaching the rig at a good rate.
- **Universe analyser** — *What is every channel doing?* Opens full-screen and lays out the whole universe as **numbered channel meters** — the channel number, a live level bar and its value — so you can read which channel is doing what at a glance and scroll the full 512. Switch between **Live**, **Min**, **Max** and **Activity** views; **Start capture** records the highest and lowest level each channel reaches over a take (that feeds the Min/Max views), and **Activity** highlights which channels are moving. Tap any channel for its detail: level, percentage, the **port it patches to**, its captured range, and how many times it has changed.
- **Flicker finder** — *Which channel is causing a flicker?* The intermittent-flicker hunter. **Arm** it and every single channel-value change on the wire is logged with the old value, the new value and how long ago it happened — so a light that flickers once every few minutes, or a desk quietly stepping a level, is **caught and timestamped** instead of leaving you guessing. It keeps logging even if you leave the tab, so an intermittent fault can be pinned to the exact channel and moment it occurred.
- **RDM universe** — *What's downstream, and can I fix it from here?* Discovers and controls every RDM fixture down the line like a dedicated RDM tester: **Scan** to find them, then read and set each fixture's **address**, **personality**, **label**, **Identify** and **sensor** readings (see §5). Scanning takes over the output line while it runs.
- **Port bus** (Developer mode only, see Setup) — a health table for the internal link to each output module (frames sent and received, CRC errors, timeouts, the module's own loop and reply timing) plus the controller's loop timing, for diagnosing a port that has stopped responding.
- **Transmit DMX** — sends a test universe from the OUT socket while no DMX is being received (greyed while a desk is live). Desk syntax on the keypad: `1 THRU 5 @ 80`, `10 + 12 @ FULL`, `1 THRU 10 - 5 @ 50`; levels are 0–100 or FULL. Each channel set is listed with its level and stays latched — across app reconnects too — until **Clear** or power-off. **Start/Stop** toggles the output; the amber banner and the **Test** pill show while transmitting. The output stops by itself when you leave Diagnostics or put the app away, so a forgotten test never holds the line; the latched channels stay for next time.

Setup — controller and housekeeping

The controller's own settings sit under **Settings** in the Controller card. Closed, one line shows what is set (name, ID, DMX hold and the current limit); tap it to open the fields. The buttons beneath stay in view either way. **DMX hold** is how many seconds the last DMX look is kept after the signal is lost, 0–60; **0** keeps it until you set something else.

Controller and Bluetooth name (one field; the Bluetooth name changes at the next restart) and **ID**, which together are what the pairing list shows, the current limit and folds, and the **CRMX by LumenRadio** card: **Unlink** and **Link** (the external antenna is always on; the Apply mode lives in port 1's Patch sheet, §Control). Two things about linking that follow from how CRMX works: a receiver is linked *by the transmitter* — put the transmitter into link mode and any unlinked receiver in range joins — so the Link button matters on the transmitter, not here; and **Unlink** makes the controller forget its transmitter so it can be claimed by another (on units with a link button on the enclosure, holding it for more than 3 seconds does the same; a short tap is ignored). The status distinguishes three things, because a CRMX link survives the transmitter being switched off: **linked — no RF** (amber) means linked but nothing is on air — transmitter off or out of range; **linked — no data** means the transmitter is present but isn't sending DMX; and **linked · N Hz** is a live stream. Then **Firmware update over Bluetooth** (§2a), **Flash to find** (identifies this controller for 5 s so you can pick it out of a rack — **it never touches the strip**: a Zone pulses all its indicator lights together, a 4our or 8ight strobes its screens), **show-file export / import** (one **.jmb** file carrying the whole rig — addresses, heads, labels, looks and scenes — reloadable on any phone or controller; a port whose Mode the receiving controller can't run keeps its own, and the app names those ports), **change PIN** (4–16 letters, numbers or symbols, no spaces; if the change never reaches the controller, the next connect still gets in with the old PIN), **Disconnect**, and **Factory reset** (everything back to as shipped: the controller unlinks its CRMX receiver first, then erases and restarts). (The rig-wide **Clear all** and its **Undo** live on the Control tab's dashboard.)

Developer mode — the small slider in the App card at the foot of Setup, remembered per phone. **Helper**, beside it and on by default, is the raised-hand icon on each card and sheet that keeps the explanatory notes out of the way until tapped; switched off, the hands go, the notes stay hidden, and the tab bar drops its names and shows larger icons alone. Developer off (the default) hides the bench detail an operator never needs: the Diagnostics *Port bus* table, the CRMX module signal lines, uptime and last-boot reason, and the script-error strip. On, it also reveals the calibration rows in the Hardware card:

- **Shunt** — which current-sense resistor the unit's port module carries. *Auto* is right for every unit as shipped (5 mΩ on every port). Only change it if the hardware has been modified: the wrong value halves or doubles every current reading *and* the trip point.
- **Cal current** — calibrates a port's current reading against a meter. Drive a steady load of at least 0.25 A, read the true current on a bench meter, type it in mA and tap *Calibrate current*; the stored gains show on the line beneath. *Reset* returns to unity.
- **Calibrate temperature** — captures each port's sensor offset against a known ambient. Leave the unit powered but dark at room temperature for ten minutes, enter the true ambient from a trusted thermometer, and every port's offset is stored. A port more than **10 °C** out is refused and flagged: that size of error is a faulty sensor, not drift, and must never be calibrated away.
- **Command** — sends a raw controller command, for bench work.
- **Controller only** in the Firmware card (§2a).

Test sequences (Setup, top of the tab) drive every port directly, overriding every source — **live DMX included** — until switched **Off**; with a desk live, the app asks before starting one. A banner appears on the Control tab while a test runs, with its own *Stop test* button, and the **Off** button lights when nothing is running so you can tell at a glance.

- **Step** — the hardware test. Every output on its own at full drive: red → green → blue → white → amber, then all five, one second each, with the Mode, curve and white balance bypassed. Identical on every port and in step across the rig, so a dead output or a weak output stage stands out immediately. An output the port's Mode does not use still lights, which shows up tape wired for a different Mode.
- **Fade** — the look test. A smooth sweep through the colours each port's Mode drives, through its curve and white balance, for judging dimming smoothness and colour blending: a rainbow on RGB, carried on through white (and amber) on RGBW and RGBWA, warm ↔ cool on a tuneable-white port, and a breathe on a dimmer port. About three seconds round the rainbow, a second longer for each extra colour; a dimmer port breathes once a second.

The wiring check runs automatically during Step. Because that test lights one output at a time, and each port measures its own current, the reading during each step *is* that output's current. After one full six-second cycle the app lists them per port:

```
Port 1: R 1.90A · G 1.60A · B 1.50A · W 1.80A · A 1.40A ✓ consistent
```

Colours legitimately differ — red's forward voltage is about a volt below green and blue — so the check only speaks up for unambiguous cases: an output drawing nothing, an output drawing markedly more than its siblings, or the two together, which means two colours are sharing a terminal (see §7). The figures stay on screen after you stop the test.

The comparison is **relative**, channel against channel, so it works on a 10 cm off-cut as readily as a 10 m run, and the load does not have to be tape. Two consequences worth knowing: a reading of only a few milliamps is reported as *no measurable current* rather than "nothing connected" — the controller cannot tell a disconnected output from a very small load — and a port is only ever **inhibited** when there is enough current for the comparison to be beyond doubt. On a small bench load a crossed pair is reported but the port stays live.

Effects on tuneable-white ports. On a CW/WW personality the effect list only offers the effects that read on a white strip (Candle, Fire, Lightning, TV Sim, Strobe, Pulse, Fluoro, Paparazzi, Explosion); the colour-only effects are hidden so you can't pick one that would do nothing. The tuneable-white personality drives the white (cool) and amber (warm) terminals.

Effects on RGBW and RGBWA ports. The effects are written in red, green and blue; the white output follows the white already in the mix (a white strobe uses it, a saturated rainbow leaves it dark) and amber stays dark under an effect. Your own white and amber levels come back when the effect is cleared.

Support. *Setup → App → Support* lets you tell us about a problem. Type what happened; the app adds a status report (versions, controller status, recent errors and commands) and hands it to your own mail app addressed to support@jmblighting.co.uk. Nothing is sent unless you send it, and *What the report contains* shows every line first. It never holds the PIN, network address or location; names, the controller ID and labels are included only if you tick the box. The privacy policy is linked from this sheet and from the Documentation card.

Demo mode. *Demo mode* on the connect screen walks the entire app — every tab, the analyser, even an RDM scan that discovers example fixtures — with no controller present. Nothing a demo user touches goes anywhere; it is for learning the app or showing it off before the hardware is in front of you.

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7. Fault handling

Ports protect themselves: over-current (hiccup ×3 then latch), bolted-short µs trip, over-temperature, ground-leak imbalance, comms-loss failsafe. Latched ports show a flashing fault banner on their screen, the fault word appears in the menu and the app, and Clear Faults / Port Reset re-arms.

Wiring fault. Separate from the electrical faults above, and raised by the controller rather than the port. Running **Step** measures each output's current one colour at a time; if one output draws roughly double while another draws nothing, two colours are almost certainly sharing a terminal. The port is **inhibited** and both the panel and the app name the pair — *P1 WIRE R+B* means the blue load is on the red terminal.

This matters because it is the one wiring mistake the port's own protection cannot see: the current still leaves and returns through the same loop, so over-current and ground-leak detection are both satisfied while one output quietly carries two channels' worth. **Clear Faults** on the panel, or *Clear wiring fault & re-enable* in the app, restores the port; if the wiring is still crossed it latches again the next time you run the test. An output with simply *nothing* connected is reported but never inhibits a port — a tuneable-white fixture legitimately has an unused output.

Thermal foldback (Setup → Thermal Fold, on by default): before a port hits its over-temp trip, it eases the output down over the last 10 °C (to a 30 % floor) so a hot fixture *rides through* instead of hard-latching mid-show. The latch stays as the real-fault backstop.

Ratings and hot environments. Every port is rated for its full 6.75 A with all ports loaded at once, at 25 °C ambient, with the air filter clean and the vents clear. In a hot rack, a sealed flight case or a sunlit truss, the controller protects itself: thermal fold eases the outputs before anything trips, and the power supply derates above its rated temperature. If the *Tfold* pill lights under the header in normal use, give the unit air. Keep the intake filter clean and never block the side slots. (The ambient at which fold first engages at full load is recorded on the bench per product and printed here once measured.)

Power link. The mains link socket (TRUE1) passes power on to the next unit. One 16 A supply takes up to **16 Zones, 4 4ours or 2 8ights** on one chain. The count allows for every unit at full output and keeps the chain inside 80 % of a 16 A breaker, so it holds whatever the limits are set to. Don't add more: the whole chain's current runs through the first unit's link.

Controller current limit (Setup → Current Limit; app → Setup, under the controller's name): the most current the whole controller may draw, in 0.5 A steps up to the product maximum — **6.75 A** on the Zone, **27 A** on the 4our, **54 A** on the 8ight. Set it to the supply you have brought (a 10 A PSU on a 4our → 10 A). **Limit Lock** freezes the value and is **on by default**: a new unit arrives locked at its product maximum, and the lock must be released before the limit can be changed, then set again. The row shows the lock with a `*` on the panel; in the app the **padlock** beside the field is the only control: one press applies the value and locks it (there is no separate Apply; Enter in the field does the same) — the field fades and only the padlock stays live until it is pressed again — so a hirer cannot raise it. **Current Fold** (on by default) is what happens

when the draw reaches the limit: every output is scaled back together — colour and balance unchanged, brightness down — until the draw sits under the limit, then eases back up. With Current Fold off the limit is only *reported* and the per-port trips remain the protection.

While either fold is scaling the outputs the panel footer reads **FOLD C** / **FOLD T** (the port screens show `CFOLD 82%` / `TFOLD 70%`), the app lights the matching pill under the header with its scale — *Cfold* in **red** for current fold, *Tfold* in **orange** for thermal — and prints the live percentage under the limit field. Tap the pill for its detail box (every pill has one).

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8. Camera & slow-motion work

Each port dims with a high-frequency PWM carrier, picked per port from the **PWM library** under **Port Settings → PWM Mode** (also in the app, and over RDM via MODULATION_FREQUENCY):

Entry	Carrier	Use it for
Smooth	8 kHz	Smoothest low-end dimming, eye-only and theatre work
Camera *(default)*	15.6 kHz	General camera work; the carrier a new unit ships on
Silent	20 kHz	Above hearing: quietest drive
Custom 1–3	2–20 kHz, your choice	Matching a camera or another fixture

All carry the full ~16-bit dimming resolution (temporal dithering makes up the difference at the higher carriers).

Custom entries. In the app, the port's *Output* sheet → PWM → *Custom…* sets up to three entries. Each holds a carrier, and any port can select it; the port keeps its own dimmer curve. The app shows the carrier the hardware will really produce. Custom entries are outside the tested set, so check the result on the strip and on camera before a show:

- below 3 kHz can show on camera and as strobing on fast movement;
- anything below 20 kHz is in the audible range, and some strips and supplies sing.

Setting up for slow-motion / high-speed capture:

- **Put every port on Camera (15.6 kHz) or Silent (20 kHz)**, and shoot a short test at your frame rate and shutter angle before the take.
- **Keep all ports on the same mode** so multiple fixtures match on camera. Use All Ports → PWM Mode, or set it once over RDM, to change the whole rig together.
- **Extreme frame rates** (multi-thousand fps) with very short exposures: shoot a quick test first. *Any* PWM dimmer can show faint banding when the exposure catches only a few carrier cycles — if you see it, widen the shutter angle or drop the frame rate slightly, or try a custom carrier. 20 kHz is the highest carrier the hardware offers.
- **Turn strobe looks off for clean capture.** The Strobe / Police effects and the FX personality's shutter channel (CH10) *deliberately* chop the output — that's the effect, not carrier flicker. Set the shutter open / effect Off.

- **Fades and colour moves are flicker-free** at the carrier; dial fade times and the dimmer curve as normal — nothing special needed for the camera.

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9. Quick reference

- Factory PIN: **0123** (entered for you; only a user-set PIN is remembered, per controller) · default addresses: sequential RGB
- DMX chart: see `docs/DMX_CHART.md`
- Start address caps at 513 − footprint so the whole personality fits the universe (FX 10ch → max 503); it drops automatically if you switch a port to a larger personality
- Universe Listen is always Off after power-up — arming is deliberate
- The app's Undo restores the looks (colour, level, effect) from before the last Clear all, on that controller only
- The app starts with no signal once it has been opened online after installing
- Supply: 24 V standard (12 V supported — match the strip voltage). All protections are current- and temperature-based, so they work identically on either rail; at 12 V each port delivers half the wattage for the same current limit
- Current limit ceiling: 6.75 A per port
- Power link: up to 16 Zones / 4 4ours / 2 8ights on one 16 A supply
- Flash to find never lights the strip; RDM discovery and Identify from a desk do — that is the RDM standard
- An RDM scan left open releases the thru line after 3 minutes idle
- Developer mode (Setup, bottom slider) holds the bench and calibration tools